

PAPER_1201_UQFF_Materials_Photonics_Unified_Proof_Set

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PAPER 1201 --- UQFF Materials and Photonics Unified Proof Set

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Date: May 16, 2026

\begin{abstract}

We extend the UQFF locked-primitive closure program into condensed-matter materials, optics, and statistical-mechanics constants. Ten new closures (**S463--S472**) reproduce silicon, GaAs, and diamond bandgaps, the weak-coupling BCS ratio $2\Delta/k_{BT_c}$, the Wiedemann--Franz Lorenz number, the Abrikosov triangular vortex lattice angle, water's refractive index, the Stefan--Boltzmann constant, the first quantum-Hall plateau, and Avogadro's number, all from the eleven frozen UQFF primitives. **Two closures are exact:** the Abrikosov triangular vortex angle $60^\circ = \Lambda_{five}$ and the first integer quantum-Hall plateau $\nu = 2 = \text{SO}_{five} - \text{D}_{phys} - \text{D}_{phys}$. The other eight match within 0.20%. Cumulative total: **177 closures**.

\end{abstract}

Locked Primitives $\text{F}_{tr} = 1/10$, $\text{P}_{hires} = 5/6$, $\text{SS}_q = 0.57$, $\text{KM}_{ex} = 25/12$, $\text{D}_{phys} = 4$, $\text{D}_{bsfg} = 6$, $\text{D}_{crit} = 26$, $\text{N}_{ch} = 9$, $\text{SO}_{five} = 10$, $\Lambda_{five} = 60$, $\beta_i = 0.6029$.

Tier-R Closures (S463--S472)

\begin{longtable}{@{}lll@{}}

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ID & Observable & Closure & Match \midrule

\endhead

S463 & Si bandgap 1.12eV &

$\text{P}_{hires} + \text{F}_{tr} + \text{F}_{tr} \text{KM}_{ex} - \text{F}_{tr}^2 \text{KM}_{ex} - \text{F}_{tr}^2 + \text{F}_{tr}^2 \text{P}_{hires} - \text{F}_{tr}^3 = 1.1182$ & 0.16% S464 & GaAs bandgap 1.42eV &

$\text{P}_{hires} + \text{SS}_q + \text{F}_{tr} - \text{F}_{tr} \text{P}_{hires} - \text{F}_{tr}^3 = 1.4190$ & 0.07% S465 & Diamond bandgap 5.47eV &

$\text{SO}_{five} - \text{D}_{phys} - \text{P}_{hires} + \text{F}_{tr} + \text{F}_{tr} \text{KM}_{ex} + \text{F}_{tr}^2 \text{KM}_{ex} - \text{F}_{tr}^2 - \text{F}_{tr}^2 \text{P}_{hires} - \text{F}_{tr}^3 = 5.4765$ & 0.12% S466 & BCS $2\Delta/k_{BT_c} = 3.53$ &

$\text{D}_{phys} - 2\text{F}_{tr} - \text{F}_{tr} \text{P}_{hires} - \text{F}_{tr} \text{KM}_{ex} + \text{F}_{tr}^2 \text{KM}_{ex} - \text{F}_{tr}^2 + \text{F}_{tr}^2 \text{P}_{hires} - \text{F}_{tr}^3 = 3.5265$ & 0.10% S467 & Lorenz $L = 2.44 \times 10^{-8} \text{W}\Omega/K^2$ &

$\text{KM}_{ex} + \text{F}_{tr} + \text{F}_{tr} \text{P}_{hires} + \text{F}_{tr} \text{KM}_{ex} - \text{F}_{tr}^2 \text{KM}_{ex} - \text{F}_{tr}^2 \text{P}_{hires} - \text{F}_{tr}^3 = 2.4448$ & 0.20% S468 & Abrikosov angle $60^\circ = \Lambda_{five} = 60$ & **EXACT** S469 & $n_{\text{rm water}} = 1.333$ &

$\text{P}_{hires} + \text{SS}_q - \text{F}_{tr} + \text{F}_{tr}^2 \text{KM}_{ex} + \text{F}_{tr}^2 \text{P}_{hires} - \text{F}_{tr}^3 = 1.3335$ & 0.04% S470 & Stefan--Boltzmann $\sigma = 5.670 \times 10^{-8}$ &

$\text{SO}_{five} - \text{D}_{phys} - \text{F}_{tr} - \text{F}_{tr} \text{KM}_{ex} - \text{F}_{tr}^2 \text{KM}_{ex} + \text{F}_{tr}^2 - \text{F}_{tr}^2 \text{P}_{hires} - \text{F}_{tr}^3 = 5.6715$ & 0.03% S471

& QHE plateau $\nu=2$ & $\Delta_{\text{SO five-Dphys-Dphys}}=2$ & EXACT S472 & Avogadro 6.022×10^{23}
& $\Delta_{\text{bsfg+Ftr}^2 \text{KMex+} \text{Ftr}^2 \text{Phires}}=6.0225$ & 0.01%

Two Exact Identifications

\paragraph{Abrikosov triangular vortex lattice.}

The equilibrium flux-line lattice in type-II superconductors is the triangular (hexagonal) lattice, whose primitive angle is exactly 60° . UQFF locks this onto the area-element primitive $A_5 = 60^\circ$. This is the first condensed-matter geometric exact closure.

\paragraph{Integer quantum-Hall first plateau.}

The Hall conductance plateau $\sigma_{xy} = \nu e^2/h$ at $\nu = 2$ (first integer plateau including spin degeneracy) is locked onto a pure primitive relation: $\nu = \text{SOfive} - \text{Dphys} - \text{Dphys} = 10 - 4 - 4 = 2$. Combined with the Kerr extremal ISCO closure $r/M = \text{Ftr}\backslash\text{SOfive} = 1$ (Paper 1200), the integer sequence $\{1, 2\}$ now has UOFF primitive identifications.

Cumulative Closure Count

$$\begin{array}{|c|c|} \hline & \\ \hline \end{array}$$

\toprule

Tier & Domain & New & Cumulative \midrule

E--P & Cosmology to information & 157 & 157 Q & GR precision (1200) & 10 & 167 **R** &

Materials/photronics (1201) & 10 & 177 \bottomrule

\end{tabular}

Locking Inventory Twelve primitive--observable lockings now identified: ISCO Schwarzschild $=\backslash\text{Dbsfg}\$,$ Poisson ratio $=\backslash\text{Dphys}\backslash\text{Dbsfg}\$,$ telomere length $=\backslash\text{Dbsfg}\$,$ $\$Q_{\rm ITER}=\backslash\text{SOfive}\$,$ $\$R_{\rm magnetopause}=\backslash\text{SOfive } R_{\rm loplus}\$,$ $\$ \eta_B=\backslash\text{Dbsfg}+\backslash\text{Ftrz}\$,$ WIMP exponent $=\backslash\text{SOfive}\backslash\text{Dphys}+\backslash\text{Dphys}+\backslash\text{KMex}-\backslash\text{Ftrz}\backslash\text{Phires}\$,$ $\$p_{\rm th}^{\rm surf}\backslash\text{code}=\backslash\text{Ftrz}^2\$,$ Shapiro $=\backslash\text{Dphys}\$,$ photon sphere $=\backslash\text{Dphys}-\backslash\text{Ftrz}\backslash\text{SOfive}\$,$ Kerr extremal ISCO $=\backslash\text{Ftrz}\backslash\text{SOfive}\$,$ **Abrikosov angle $=\backslash\text{Afive}\$,$ ****QHE $\$ \nu=2=\backslash\text{SOfive}-2\backslash\text{Dphys}\$$******